

RESPIRATORY SYSTEM

NCLEX STUDY GUIDE



Organized for study and printing | 16 content pages

Respiratory system

- Inspection
- Palpation
- Percussion
- Auscultation

Diagnostic tests**Sputum Specimen.** (Tuberculosis & Pneumonia)

- Obtain early in the morning.
- Sterile specimen (can be from suctioning or expectoration)
- Instruct the client to rinse mouth with water before collection.
- Obtain 15 ml of sputum.
- UAP collecting the specimen.

Board: Specific Test to diagnostic TB is sputum specimen (3 collection).

- ◇ Mucus trap is use take the sample



◇

Laryngoscopy and Bronchoscopy. (Invasive procedure).

Direct visual examination of larynx- trachea and bronchi.

1. Informed Consent.
 2. Chest X-Ray before.
 3. NPO Status.
 4. Coagulation stadium before.
 5. Remove denture and eyeglasses.
 6. At the bedside Emesis basin, suction equipment and resuscitation equipment.
- After procedure Check gag reflex.
 - Complication= Pneumothorax.

Early sign of Pneumothorax Call Dr. Immediately.

- Skin crepitus (shoulder, chest)
- Decreasing or absent breath sound unilateral.

Thoracentesis (Invasive)

Removal of fluid or air from the pleural space.

- Inform consent.
- Chest X-Ray.
- Coagulation studies.

Position

Sitting up right with arms and shoulder supported in a bedside table.

Complication

1. Pneumothorax.
2. If it leaks, call the Dr.
3. Embolism
4. Pulmonary Edema

Nota: No se llama al Dr. Cuando es liquido en la barriga (ex: ascitis).

Pulmonary Function Test

1. Lung Mechanic
2. Gas Exchange
3. Lung volumen
4. Acid base disturbance
5. Arterial blood gas level.

Nota: en el board preguntan para que sirve este test, y te mencionan varias en una oración: casi siempre se escoge la más completa.

Preparation for the test

- Client has to void first
- Removed denture
- Use loose clothes
- Refrain from smoking and heavily eating meals 4-6 hours before.
- Withhold bronchodilators 6-12 hours before.

Skin Test (Subcutaneous)

- Use a skin siter that is free of excessive body hair and dermatitis.
- Apply the injection to the upper 1/3 of the inner surface of the left arm. SQ
- Circle and mark the injection.
- Avoid scratching or washing.
- Document the day and time
- Read 24-72 hours.
- If blood is present won't be accurate reading.

Chest Injury

1. Ribs fractures
2. Flai chest
3. Pneumothorax
 - ◇ Pneumothorax spontaneous
 - ◇ Open Pneumothorax
 - ◇ Tension Pneumothorax

1. Ribs fractures

- ✓ Less than two Ribs
- ✓ Result from blunt trauma

Assessment

1. Pain at the injury site increased with inspiration.
2. Tenderness at the site.
3. Shallow respiration = (Respiratory Acidosis)
4. Client splint chest.
5. Fx are noted on the X-Ray.

Complication: Pneumothorax

Intervention (chest injury)

1. Ribs usually unite spontaneously.
2. Place client in fowler position.

3. Pain medication (con medicamento que termine en **caine**, que son los anestésicos).
4. Prepare for an intercostal Nerve Block

Flail Chest RED CODE

- More than 2-3 consecutive Rib Fracture.
- Result direct blunt chest trauma.
- ICU admission.

Assessment

1. Paradoxical Respiration.

- ✓ Inward movement of the segment of the thorax during inspiration with outward movement during expiration.
- 1. Severe chest pain
- 2. SOB, Cyanosis, tachycardia, hypotension, shallow respiration, decreased breath sound bilateral.
- 3. Pain

Note: Pulsus paradoxus:

- ✓ Status Asthmatics
- ✓ Cardiac Tamponade
- ✓ ↓ systolic + 10 mmhg at the end of the expiration.

Intervention

- Semi fowler to fowler
- O2

Mechanical Ventilation

Modes of Ventilation		
Controlled ▫ Used for clients who cannot initiate respiratory effort. ▫ Last used	Assisted-Control -Client can initiate breathing effort ventilate is programmed to responded. -The client inspiratory effort. -More risk of Respiratory Alkalosis	SIMV (synchronized intermittent mandatory ventilator). -Weaning mode allows the client to breath spontaneously at her or his own rate between ventilator breaths -Sedation holidays

Intervention

1. Check client first.
2. Ventilate manually with Ambu-bag until the cause is determinate.

Low Pressure Alarm	High Pressure Alarm
➤ Machine 1. Circuit leaks 2. Airway leaks 3. Chest tube leaks 4. Pt disconnected 5. Pt stop spontaneously breathing 6. Cuff deflated 7. Leak in ventilator 8. Power off	➤ Patient 1. Airway secretions 2. Whizzing and Bronchospasm. 3. Client coughing 4. Gagging 5. Pt attending to talk 6. Client biting an oral ET Tube. 7. Client fighting with ventilator. 8. Endotracheal Tube displaced.

- | | |
|---|--|
| 9. Tube disconnected from ventilator
10. Ventilator malfunction. | |
|---|--|

Pneumothorax

- Accumulation of atmospheric Air in Pleural space which results in an increase of intrathoracic Pressure.

Types

- Spontaneous (smokers).
- Open Pneumothorax = Trauma (sucking sound).
- Tension Pneumothorax = Tracheal deviation to opposite site.

Assessment

- Decreased or Absent breath sounds unilateral.
- Skin crepitus = subcutaneous emphysema.
- Cyanosis, SOB, Tachycardia, Hypotension, hemoptysis, chest pain.

Intervention

- O2.
- Fowler position.
- If open = Apply a three-side dressing on an open chest wound to prevent tension Pneumothorax.

Chest Tube

Suction Control Chamber	Water seal chamber		Drainage Collection Chamber 70-100 ml
20 ml of water to function. - Refill if is less than 20ml. - Continuous gently bubbling. - If vigorous intermittent bubbling. Call the Dr.	Intermittent bubbling. ↑with inspiration. ↓with expiration. If continuous bubbling = Air leakage = clamp the tube to check de donde esta saliendolo.	Tiding or fluctuation. ↑with inspiration ↓with expiration If stop tiding or fluctuation: 1. Obstruction or kink of the tubing 2. Lung Re-expansion Auscultate BS X-Ray. 3 Dependent loop 4 Machine malfunction	Monitor every 4hrs the drainage with piece of tape. If more than 100ml of blood or bright red drainage = Call the Dr. Except for the first 24hrs 500-600 ml Call the Dr. if bright, red drainage. Dark red= old is expected and no required follow up.

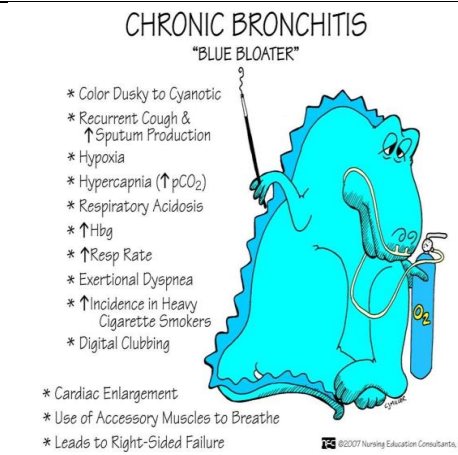
- If chest tube become dislodged
 - ◇ Apply pressure over the insertion site with three sides dressing.
- If chest tube become disconnected from drainage system.
 - ◇ Cut the tip of contaminated tubing, immerse in 2 cm of sterile water, until system can be reestablished.

Chest tube removal

- Gathering supplies (dressing or petroleum jelly)
- Explain the procedure

3. Give medication (15 minutes before)
4. Cut the suture
5. Holding his/her breath or bearing down
6. Remove the tube.
7. Cover with petroleum jelly and dressing.
8. Document

COPD Chronic Obstructive Pulmonary Disease
(use venturi mask or nasal canula)

EMPHYSEMA (Pink panther)	CHRONIC BRONCHITIS
1. Pink Puffer 2. Under weight 3. Alveolar membrane break down 4. Minimal cough 5. Barrel chest = AP Diameter = Follow up = 1:1 (2:1 = Ok) - L/E 2:1 = ok Mature Lungs 6. History of alfa 1 antitripsine deficiency	1. Blue Bloater 2. Overweight due to edema 3. Inflammation of Bronchi and excess mucus. 4. Cough with Increased mucus production = persistent X 3 months to at least 2 consecutive years. 5. Hemoptysis.
 * ↑ CO₂ RETENTION (PINK) * NO CYANOSIS * PURSE LIP BREATHING * DYSPNEA * INEFFECTIVE COUGH * HYPERRESONANCE ON CHEST PERCUSSION * ORTHOPNEIC * BARREL CHEST * EXERTIONAL DYSPNEA * PROLONGED EXPIRATORY TIME * SPEAKS IN SHORT JERKY SENTENCES * ANXIOUS * USE OF ACCESSORY MUSCLES TO BREATHE * THIN APPEARANCE * LEADS TO RIGHT SIDED ♥ FAILURE	 * Color Dusky to Cyanotic * Recurrent Cough & ↑ Sputum Production * Hypoxia * Hypercapnia (↑ pCO₂) * Respiratory Acidosis * ↑ Hgb * ↑ Resp Rate * Exertional Dyspnea * ↑ Incidence in Heavy Cigarette Smokers * Digital Clubbing * Cardiac Enlargement * Use of Accessory Muscles to Breathe * Leads to Right-Sided Failure
COPD Continuos Both 6. Expiratory Dyspnea. 7. Wheezing 8. Clubbing Nails 9. Polycythemia (increase hemoglobin to compensate) 10. Cor Pulmonale (Lung hypertension) = right side CHF S/S = CHF right • Edema • Hepatomegalia (liver aumentado de tamaño) • Jugular vein distention. • Hepato jugular reflujo • Ascitis Nota: en COPD nunca se usa el spirometer	

Treatment. (COPD)

1. Smoking Cessation.

2. Pursed lip breathing technique to prevent air trapping.

- Breathe in as if smelling flower through nose in 2 times.
- Do not puff out cheeks.
- Tuck lip to offer resistance exhaling.
- Use abdominal muscle to breath out or use a splinting (apretando la barriga)
- Count until 4 when breathing out through mouth as blow candle.

1. Huff Cough

- **To remove secretions**
- Client should take 3 or 4 deep breaths using pursed lip and diaphragmatic breathing.
- Leaning forward and the client should cough 3 or 4 times during exhalation.



2. Risk for pneumonia

- Influenza vaccine
- Pneumococcal vaccine

3. Nutrition

- High Calories
 - High protein
 - Good fat (Peanut butter sandwich).
 - **Low carbohydrate** (todas las O2 diseases llevan low carbohydrate)
 - Supplement (Ensure or milk shake between meals).
 - No liquid after meals
 - Fatigue – rest before meals, bronchodilator before meals.
 - O2 during meals.
 - Avoid gas forming food
- Broccoli
 - Avocado
- beans
 - lettuce
4. Low O2 = 2L/min

5. Medications

1. Anticholinergics	Solo abre:	Cierra choro:
Ipratropium Tiotropium Nota: No in clients with peanuts allergies	Eyes = Miadrisis Hear t=Tachycardia Lungs = Bronchodilators	↓ Tear ↓ Salivation ↓ Constipation ↓ Urinary retention ↓ Sweating
2. B- Agonist		Notal: Los terol suben todo
Albuterol. (Acute) Salmeterol Bambuterol	Tachycardia ↑ BP ↑ Glucose level	
3. Inhaled Glucocorticoides		Also, Cushing medication
Fluticasone → Budesonide		

Triancinolone	Oral thrush-mouth wash after to prevent side effect = candidacies	
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Note: All stomatitis is treated with nystatin except methotrexate in chemo that is treated with Leucovorin (folic acid).

Masks

1. Nasal Canula O₂ Delivered = 1-6L/mint (F₁O₂ =24-44%) <u>Nursing Consideration</u> 1. Can dislodge easily 2. Allow client to eat and speak 3. Use water soluble lubricant to prevent nasal mucosa dryness 4. Assessing skin integrity.	
2. Simple Face Mask O₂ Delivered = 5-8 L/mint (F₁O₂ =40-50%) <u>Nursing Consideration</u> 1. Interfere with eating and talking 2. Assess for claustrophobia 3. Avoid if client is vomiting 4. Remove saliva and mucus from the mask. 5. Pad straps in front the ear to prevent skin breakdown.	
3. Venturi Mask O₂ Delivered = 4-10L/mint (F₁O₂= 24-55%) ✓ Choice for COPD ✓ Delivered more exact amount of O₂ <u>Nursing Consideration</u> 1. Keep the air entrapment port for the adapter open and uncover. 2. Keep the mask snug on the face and ensure flexible tube is free for kinking or obstruction.	
4. Partial Rebreather Mask O₂ Delivered = 6-15L/mint (F₁O₂= 70-90%) ✓ Never on COPD ✓ Choice for Respiratory Alkalosis ✓ Panic Attack ✓ Labor <u>Nursing Consideration</u> 1. Adjust flow rate to keep the reservoir bag 2/3 full during inspiration. 2. Keep mask snug on face. 3. Make sure the reservoir bag does not twist or kink. 4. Remove mucus and saliva from mask.	

5. Non-Rebreather Mask F1 O₂ = 60-100% ✓ Use in emergency situation ✓ Before Mechanical Ventilation <u>Nursing Consideration</u> 1. Adjust el flow rate to keep the reservoir bag inflated 2/3. 2. Keep mask snug on the face. 3. Remove mucus and saliva. 4. Emotional support for claustrophobia. 5. Ensure that the valves and flaps are intact and functional. <u>Valves:</u> • Open with expiration • Close with inspiration. ✓ Reservoir bag do not twist or kink.	
6. Tracheostomy Collar and T-bar or T piece. ○ Special Adaptor ○ Add Humidifier <u>Nursing Intervention</u> 1. Empty condensation tube to prevent aspiration at least every 4 hours. 2. Keep the exhalation port in a T-piece open or uncovered. 3. Position the T-piece so that it does not pull on the tracheostomy.	

Asthma

- Chronic Inflammatory disorder of the airway that causes obstruction, inflammation.
- Hyperresponsive to a variety of stimuli or triggers.

Triggers Factors

1. Allergens inhalation:
 - Dander (caspa de perros).
 - House dust mist.
 - Air pollutants
 - Smoking
 - Perfum
 - Aerosol Spray
2. Occupational Exposure
 - Paint
 - Laundry detergent
3. Food
 - Wine
 - Beer
 - Shellfish
 - Processed Potatoes

4. Intrinsic
 - Menstruation
 - Gerd

5. Medication
 - ASA
 - NSAID
 - B- blockers

Nota: **AMBA** = Safe in **A**sthma

Assessment

1. SOB. (Nasal Flaring)
 2. Chest Tightness
 3. Expiratory dyspnea
 4. Wheezing on auscultation
 5. Dry Cough
 6. R-R > 20 = SOB
- Expected

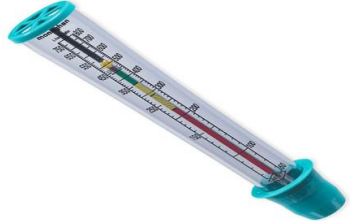
Severe Asthma Attack and Status Asthmatics

- ✓ R-R > 30 = Call Dr.
- ✓ Restlessness, Anxiety, Increased cyanosis.
- ✓ Pulsus paradoxus (same in cardiac tamponade) and paradoxical respiration (flails chest)
- Decrease more than 10 mmhg BP Systolic at the end of inspiration (equal al Cardiac Tamponade).

➤ Ominous sign of impending respiratory failure

- Sudden relieve of wheezing (silent Chest)
- Red Code = Emergency

➤ Diagnostic (**Peak Flow Meter**)

1. Set the peak flow at zero. 2. Use standing position without leaning forward or support yourself on anything. 3. Take a deep breath 4. Place a mouth-piece in your mouth (wrap your lips tightly around it). 5. Blow your breath as fast as you can. 6. Reset and preform the test 2 more times		
Green Zone + 80% = Ok Document	Yellow Zone 50-79% = Albuterol + Dr. Appt.	
		Red Zone < 50% = Emergency Intervention. 1. Albuterol + Call 911.

Nota: Peak flow meter = blow
 Spirometer = Suck

Medication

1. Acute Asthma
 - Albuterol

SABA

- Terbutaline
- Aminophylline/Theophylline

2. Long Term Chronic Asthma

- Anticholinergic: Tiotropium/ Ipratropium S/E mydriasis, bronchodilation, tachycardias.
- Aminophylline/Theophylline

To prevent:

- Montelukast/ Zafirlukast (leukotriene modifier hepatotoxic) 1h before or 2h after meals, Liver toxicity.
- Cromoglicate
- Inhaled and parenteral steroids.

Pulmonary Embolism**Causes:**

1. Thromboembolism = DVT
 - ◇ Obeso/Pregnant
 - ◇ Long travel
 - ◇ Surgery
 - ◇ Immobility
 - ◇ Oral contraceptive
2. Fat Embolism = Long bone fracture (hip, femur)
 - ◇ S/S = 48/72h petechias chest or neck, thorax and hands
3. Air embolism
 - IV Line/ Central Line.
4. Amniotic fluid Embolism = Pregnant, after amniocentesis.
5. Septic Embolism = Endocarditis.

Assessment

1. Sudden Collapse.
2. SOB, tachycardia, tachypnea, Cyanosis, Chest pain, Hemoptysis
3. LOC alteration, confusion.
4. NVD (Neck vein distention)
 - Increased CPV
 - Increased PCWP (presion arterial pulmonar)

Diagnostic

D-Dimer – positive= Thrombosis = hay coagulo, nose sabe donde.

VQ Scan = Confirmatory Diagnostic, have Dye.

Treatment

1. ICU admission
2. Thrombolytic therapy if DVT
3. Anticoagulante
4. Vena Cava Filter
5. Increased HOB 30-45

- | |
|---|
| 6. In case of FAT Embolism
▪ Prone position 1 st Option.
7. Air Embolism
▪ Trendelenburg left side if not of increase ICP |
|---|

Pneumonia

- Inflammation of the lung parenchyma that is caused more frequently by microorganisms. Ex: Bacterial, virus

Risk Factor

1. Splenectomy (#1) (pneumococcal vaccine)
2. Sick cell Anemia
3. Thalassemia
4. Age > 60
5. Bed rest
6. Alcoholism
7. Head injury
8. COPD
9. Stroke
10. Seizure
11. Mechanical ventilator
12. HIV = Jiroveci or pneumocystis Caroni (Standard precaution)

Types	
1. Community Acquired Pneumonia	2. Nosocomial or Hospital Acquired Pneumonia
Community and during first 48h of hospitalization. Streptococcus Pneumonia (Pneumococcus) Droplet Precaution	More than 48 hrs. after hospital admission ✓ Pseudomonal = bacteria ✓ Klebsiella ✓ Enterobacter – E. Coli ✓ Proteus

Droplet	Standard
➤ Pneumococcal ➤ Walking Pneumonia (Atypical) ➤ Haemophiles Pneumonia (epiglottitis)(meningitis) ➤ Pneumonic Plague	➤ Jirovecy or pneumocystis carinii (HIV)(AIDS) ➤ Legionary Disease (AC) warm stagnant water ➤ Sarcoidosis (bacteria, virus or chemicals) ➤ Aspergillosis (medio ambient) ➤ Histoplasmosis (Muerciologo) ➤ Pneumonic Anthrax.

Assessment

1. Fever with chills
2. Pleuritic chest pain
3. Crackles or **rales** on lungs auscultation (Unilateral).
4. Dullness or Percussion
5. Tacti Fremitus Increased. (Say 99)

Nota; gallinas comen bien en tierra, vuelan mal en aire y se ahogan en agua.

1. SOB
2. Rusty Colored Sputum or mucopurulent pneumococcal.

Note: pink frothy sputum = acute pulmonary edema

Diagnostic Pneumonia

1. Chest X-Ray
2. Confirmatory Diagnostic (Sputum)

Treatment

1. O₂
2. Semi fowler to fowler 30-45 or over unaffected side.
3. Encourage coughing and deep breathing.
4. Chest physiotherapy
 - o Administer Albuterol 30 minutes before (same for Cystic Fibrosis), also can administer Acetylcysteine.
1. Antipyretic (fever)
2. Antibiotic
3. Diet
 - o High Calories
 - o High proteins
 - o Low Carbohydrates
 - o Encourage Fluid intake

Ventilator Associate Pneumonia

Board- SATA

1. HOB 30°-45° (if separated choose 30°)
2. Oral Care with normal saline every 4 hours or with chlorhexidine twice daily.
3. Daily Evaluation for extubating (weaning)
4. Sedation holiday (daily interruption of sedative infusion allowing the client to wake).
5. Cuff Pressure Assessment = twice daily (20 = normal)
6. Hand washing

Tuberculosis

- Highly communicable disease caused by Lethargy mycobacterium tuberculosis (Cosh bacilli)
- Multidrug resistant (full tto = 6-9 months)

Risk factor TB

1. Children less than 5 years.
2. Drinking unpasteurized milk.
3. Homeless
4. Lower socioeconomic groups, refuge groups.
5. Living in crowded area (long-term care facilities, mental health facilities)
6. Elderly > 60
7. Immunocompromised
 - ✓ HIV
 - ✓ Chemotherapy
1. Malnutrition

2. Alcohol and IV drug abuser

Note: Client received Bacilli-Calmette-Guerin (BCG) vaccine = PPD positive = Do not confirm Active TB.

Assessment TB

1. Night Sweat
2. Low grade of fever + chills.
3. Persistent cough with mucoid or mucus-purulent sputum which occasionally streaked with blood.
4. Fatigue
5. Lethargy
6. Anorexia = weight loss
7. Chest tightness and aching pain

Diagnostic

1. Quanti FERON TB “Gold Test”
 - ✓ Blood analysis Test.
 - ✓ A sensitive and rapid test.
 - ✓ Result available in 24 hours **but no confirmatory.**
2. Sputum Cultures = Confirmatory diagnostic (3 muestras)
3. X-Ray
4. Mantoux skin test (PPD)
 - ✓ Purified protein derivate containing 5 tuberculin units is administered intradermal in the forearm.
 - ✓ Read 48-72h
 - 10 mn of induration
 - 5 mn just for immunocompromised and recent exposure.
 - If positive PPD once = X-Ray next time (lesion seen in the apex)
 - Before PPD avoid immunosuppressive therapy = steroids

1. Intervention

- The hospitalized client
 - Airborne precaution = Negative Pressure room.
 - **At least 6-12 exchanges of fresh air per hour.**
 - Nurses use N-95 before entering the room, remove after getting out and door closed.
 - When the client is going outside the room, put surgical mask.

Note: N-95 or Hepa filter, particulate mask, special mask, fits mask.

2. Client education

- At home after 3-4 weeks after treatment.
 - Return to work after 3 negative sputum culture.
 - Cough Etiquette:
 - Instruct client to cover mouth and nose when coughing or sneezing.
 - Put used tissues into an individual plastic bag.
 - Sleep alone, do not share food utensils.

Diet

- ✓ High in Iron.
- ✓ High in protein

- ✓ High in Vit C
- ✓ Low Carbohydrate

Medication TB (RESPI) = first line

1. Rifampin
 - Orange discoloration of body secretion: Urine, Tears = **Expected** = Avoid contact lenses
 - Liver toxicity
 - Decreased effect oral contraceptive
2. Ethambutol
 - ◇ Optic Neuritis
 - - No in children less than 13 years old
 - - No in client with eyes problems
 - Eye examination during treatment and before
 - No in eye problems clients
3. Streptomycin
 - Nephrotoxicity Increase BUN and creatinine
 - Ototoxicity = tinnitus
4. Pyrazinamide
 - Increased Uric Acid (no in gout)
 - No with thiazide diuretics bc increase uric acid.
5. **Isoniazide** (profilactic and tto) (si un miembro de la pareja lo tuvo, el otro también lleva este tto también)

Side effect

- ✓ Peripheral neuropathy gives Vit B6 to decrease the effect of peripheral neuropathy.
- ✓ Liver toxicity

Second line:

- ◇ Ciprofloxacin
- ◇ Aminoglycosides (Micin)

Acute Respiratory Distress Syndrome (ARDS)

- Prone position
- ICU Admission
- A form of acute respiratory failure that occurs as a complication of other conditions.

Causes:

- ✓ Sepsis
- ✓ Fluid overload
- ✓ Shock
- ✓ Trauma
- ✓ Neurologic Injury
- ✓ Burn
- ✓ DIC
- ✓ Drug ingestion
- ✓ Aspiration

Assessment

1. **Refractory hypoxemia** high concentration of O₂ or do not improve with 100% O₂. (does not get better with tto = nonrebreather mask)
2. Hypoxia despite
3. Tachypnea
4. Pulmonary edema but not cardiogenic.
5. Decreased breath sound
6. Respiratory Acidosis
7. Pulmonary infiltrates = **“ground glass”** appearance visible. X-Ray.
8. PO₂ <60 %

Intervention

1. Fluid Restriction
2. Mechanical Ventilation
3. Prone Position.
4. Diuretics

Legionnaires Diseases

- Acute bacterial infection
- Source of the organism included contaminated cooling tower water and warm stagnant water supply including water vaporizer whirlpools spas, shower, AC
- **Standard precaution**

Assessment

- Like to pneumonia
- Increase fever
- Chills, muscle aches

Treatment

- Antibiotic
- Teaching about water accumulation

Histoplasmosis

- Pulmonary fungal infection
- Caused by spores.
- Are usually found in bird droppings (caca)
- **Standard precaution.**

Assessment

- Like to pneumonia
- Positive skin test for Histoplasmosis + Hepato-Splenomegaly.

Treatment

1. O₂
2. Antifungal (**amphotericin B**) (único antifungal que se pone IV) S/E nephrotoxicity and liver toxicity. No with zidigoxin bc decrease K.
3. Anthistaminic
4. Stereoids.
5. Semifowler position
6. Instruct client to spray the floor with water before sweeping barn and chicken coops.

Sarcoidosis

- Presence of Epithelioid cell tubercles in the lung.
- Associated with Epstein- Barr = (mononucleosis) (standard precaution)
- Black young adult.

Assessment

1. Night sweat
2. Fever
3. Weight loss
4. Cough and SOB
5. Skin nodules
6. Polyarthrititis
7. Test positive Kvein = (Diagnostic)

Treatment

- Steroids

Occupational Lung Disease

- Caused by exposure to environmental or occupational
- ✓ Silicosis or coal miner's (minero)
 - ✓ Asbestosis (cemento)
 - ✓ Talcosis (polvo)
 - ✓ Berylliosis

Standard precaution – risk for COPD